

Statistical estimation of Optimal transport maps

SC Team Seminar

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CNRS - INP - UT3 - UT1 - UT2J



- 1 Introduction
- 2 Construction of the estimator
- 3 Convergence of our estimator
- 4 Discussion

From the Monge to the Kantorovich problem

Monge Problem:

$$\inf_{\substack{T: X \rightarrow Y \\ T_{\#}\mu = \nu}} \int_X \|x - T(x)\|^2 d\mu(x)$$

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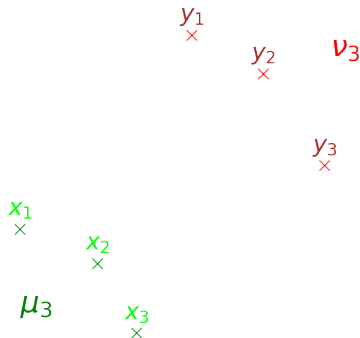


Figure: 3 source atoms to 3 target atoms

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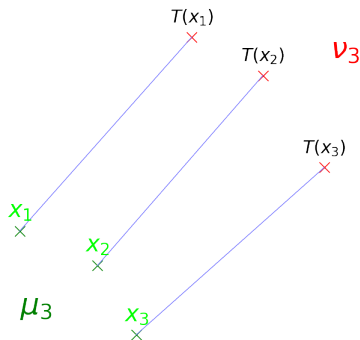


Figure: Optimal transport map $T : \{x_1, x_2, x_3\} \rightarrow \{y_1, y_2, y_3\}$

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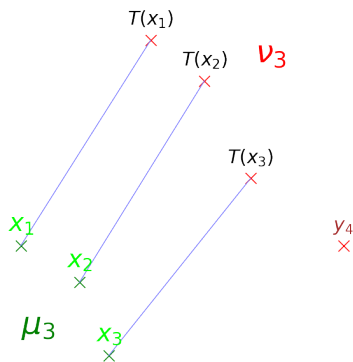


Figure: 3 source atoms to 4 target atoms

From the Monge to the Kantorovich problem

Kantorovich's relaxation:

$$\inf_{\gamma \in \Pi(\mu, \nu)} \int_{X \times Y} \|x - y\|^2 d\gamma(x, y) \quad (:= W_2^2(\mu, \nu))$$

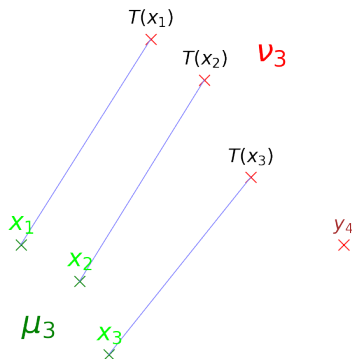


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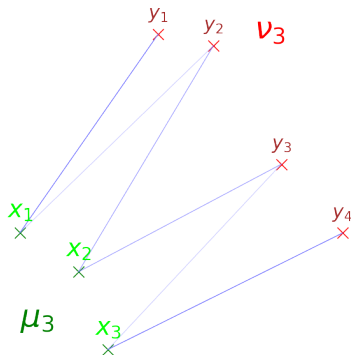


Figure: Optimal probability coupling $\gamma \in \Pi(\mu_3, \nu_3)$

A sufficient condition: Brenier's Theorem

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When does (MP) have a solution?

Theorem (Brenier)

If μ is absolutely continuous with respect to Lebesgue then there exists a solution to (MP). Moreover there exists a convex function φ such that for μ -almost all $x \in X$:

$$T^*(x) = \nabla \varphi(x)$$

Outline of the problem

- μ is an absolutely continuous population measure known only through samples.

$$\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n \delta_{X_i} \quad X_1, \dots, X_n \stackrel{iid}{\sim} \mu$$

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- Brenier's Theorem $\implies \exists T^*$ such that $\nu = T^*_{\#} \mu$.
- In general no OT map for $\hat{\mu}_n, \hat{\nu}_m$.
- How can we estimate T^* from $\hat{\mu}_n$ and $\hat{\nu}_m$?

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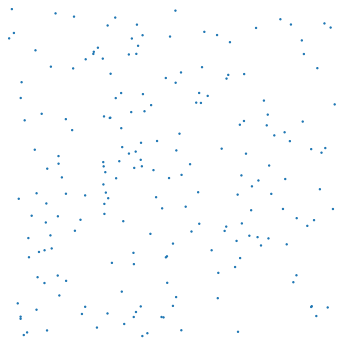
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We leverage this property to build a statistical estimator for T^* :

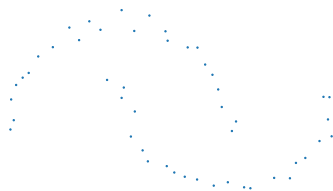
$$\hat{T}_{n,m} \in \partial \hat{\varphi}_{n,m}$$

Where $\hat{\varphi}_{n,m}$ is a suitably chosen **Brenier potential** for $OT(\mu_n, \nu_m)$.

Illustration of the estimator

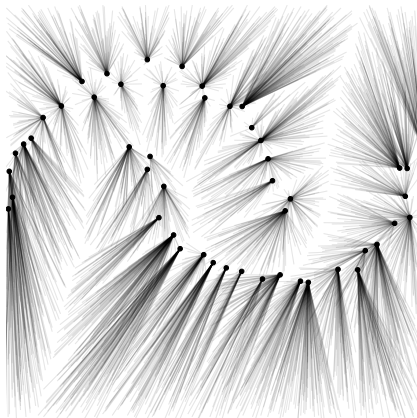


Source measure



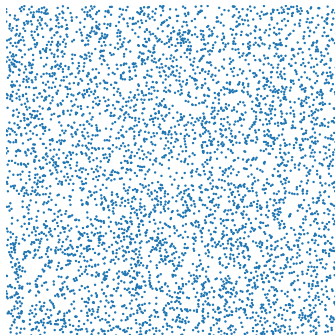
Target measure

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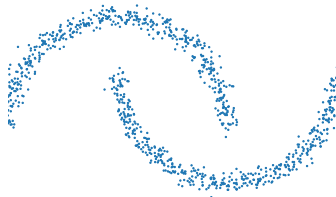


Monge map representation

Illustration of the estimator

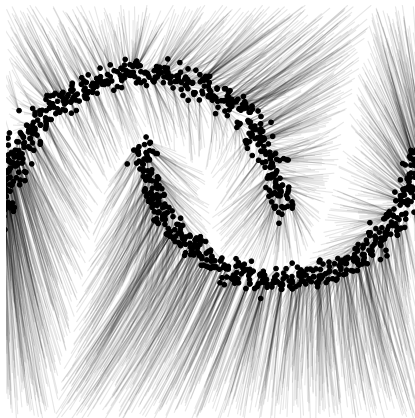


Source measure (Large sample size)



Target measure (Large sample size)

Illustration of the estimator



Monge map representation (Large sample size)

Some convex analysis

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We also recall the following (classical) property:

Subgradient of max-functions: If $f(x) = \max\{f_i(x), i \in I\}$, where all f_i are $\mathcal{C}^1$, then

$$\partial f(x) = \text{Conv}\{\nabla f_j(x) \mid f_j(x) = \max_i f_i(x)\}.$$

Duality Theorem

Theorem (Duality for quadratic optimal transport)

μ, ν be compactly supported in $B(0, R)$. Consider

$$\inf_{\gamma \in \Pi(\mu, \nu)} \int \frac{1}{2} \|x - y\|^2 d\gamma(x, y) \quad (\text{KP})$$

$$\inf_{\varphi \in \Gamma_0(\mathbb{R}^d)} \int_{\mathbb{R}^d} \varphi d\mu + \int_{\mathbb{R}^d} \varphi^* d\nu. \quad (\text{DP})$$

Then the following are equivalent. For any $\gamma \in \Pi(\mu, \nu)$ and $\varphi \in \Gamma_0(\mathbb{R}^d)$:

- ① γ is optimal for (KP) and φ is optimal for (DP).
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Moreover:

Lemma $\implies \varphi, \varphi^*$ may be chosen $2R$ -Lipschitz on $B(0, R)$.

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A necessary first-step is discrete transport.

The discrete problem

Recall the Kantorovich problem:

$$\inf_{\gamma \in \Pi(\mu, \nu)} \int_{X \times Y} \|x - y\|^2 d\gamma(x, y) \text{ (KP)}$$

The discrete problem

In the discrete case, if $\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n \delta_{X_i}$ and $\hat{\nu}_m = \frac{1}{m} \sum_{i=1}^m \delta_{Y_i}$, (KP) becomes (KP-discr):

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$$\min \left\{ \langle \hat{C}, \gamma_{n,m} \rangle \mid \gamma_{n,m} \in \mathbb{R}_+^{n \times m}, \sum_{j=1}^m \gamma_{n,m}(i,j) = \frac{1}{n}, \sum_{i=1}^n \gamma_{n,m}(i,j) = \frac{1}{m} \right\}$$

where $\hat{C}$ is the cost matrix with entries $\frac{1}{2} \|X_i - Y_j\|^2$.

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where $\hat{C}$ is the cost matrix with entries $\frac{1}{2} \|X_i - Y_j\|^2$. Once again, there is a duality formula for (KP-discr):

$$\max \left\{ \frac{1}{n} \sum_{i=1}^n f_i + \frac{1}{m} \sum_{j=1}^m g_j \mid f_i + g_j \leq \hat{C}_{i,j}, (f, g) \in \mathbb{R}^n \times \mathbb{R}^m \right\}.$$

The Brenier estimator

Saturating the constraint in (KP-discr) we obtain

Lemma (Complementary slackness)

Let $\hat{\gamma}_{n,m}$ be an optimal coupling matrix in (KP-disc) (f, g) be optimal dual variables for the dual formulation of (KP-discr). Then

$$\hat{\gamma}_{n,m}(i, j) > 0$$

$$\implies f_i \equiv \min_{k=1, \dots, m} \frac{1}{2} \|X_i - Y_k\|^2 - g_k = \frac{1}{2} \|X_i - Y_j\|^2 - g_j$$

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By the convex analysis result on the subdifferential of max-functions:

$$\partial \hat{\varphi}_g(x) = \text{Conv} \left\{ Y_i \mid i \in \underset{i=1, \dots, m}{\operatorname{argmax}} \langle x, Y_i \rangle + g_i - \frac{\|Y_i\|^2}{2} \right\}$$

So $Y_j \in \partial \hat{\varphi}_g(X_i)$.

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This result is the discrete analog of $\text{supp } \gamma \subset \text{graph}(\partial \varphi)$. By the duality Theorem $\hat{\varphi}_g$ is thus an optimal Brenier potential.

The transport map estimator

- $\hat{\varphi}_g(x) := \max_{k=1,\dots,m} \langle x, Y_k \rangle + g_k - \frac{1}{2} \|Y_k\|^2$
- $\partial \hat{\varphi}_g(x) = \text{Conv} \{ Y_i \mid i \in I(x) \}$ where
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Remark

This estimator is not feasible, nor optimal for the empirical problem. However it is asymptotically (as we will show), while being natural (the gradient of a convex function) and straight forward to implement (requires only the computation of a vector in the linear program (KP-discr)).

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Assumption

- μ is absolutely continuous, compactly supported with upper bounded density by $M > 0$.
- ν is compactly supported.
- $X_1, \dots, X_n \stackrel{iid}{\sim} \mu, \hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n \delta_{X_i}$
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Brenier's Theorem $\implies \exists \bar{\varphi}, \nu = \nabla \bar{\varphi}_\# \mu$.

Assumption

- μ is absolutely continuous, compactly supported with upper bounded density by $M > 0$.
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Objective: Provide convergence rates for

$$\mathbb{E} \left[\left\| \hat{T}_{n,m} - \nabla \bar{\varphi} \right\|_{L^1(\mu)} \right].$$

A stability result for Brenier potentials

Theorem

There is $C_{R,d,M} > 0$ such that for any $\bar{\varphi}, \phi \in \Gamma_0(\mathbb{R}^d)$ such that $\bar{\varphi}, \phi$ and $\bar{\varphi}^, \phi^*$ are $2R$ -Lipschitz on $B(0, 2R)$ and $\nabla \bar{\varphi}_\# \mu = \nu$:*

$$\int \|\nabla \bar{\varphi} - \nabla \phi\| d\mu \leq C_{R,d,M} \left(\int \phi d\mu + \int \phi^* d\nu - \int \bar{\varphi} d\mu - \int \bar{\varphi}^* d\nu \right)^{\frac{1}{4}}.$$

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Characterization of W_1 in terms of Lipschitz functions yields

Corollary

Under the same hypothesis on $\bar{\varphi}$:

$$\int \|\nabla \bar{\varphi} - \hat{T}_{n,m}\| d\mu \leq C_{R,d,M} (W_1(\mu, \hat{\mu}_n) + W_1(\nu, \hat{\nu}_m))^{\frac{1}{4}}.$$

Theorem (Main Theorem)

Let μ be an absolutely continuous measure with upper-bounded density by $M > 0$ and ν any measure, both supported in $B(0, R)$. Then

$$\mathbb{E} \left[\|\hat{T}_{n,m} - \nabla \bar{\varphi}\|_{L^1(\mu)} \right] \leq C_{R,d,M} \left(n^{-\frac{1}{4d}} + m^{-\frac{1}{4d}} \right).$$

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And for discrete $\nu_m = \sum_{i=1}^m b_i \delta_{y_i}$ (not random):

Theorem (Main Theorem-semi-discrete)

$$\mathbb{E} \left[\|\hat{T}_n - \nabla \bar{\varphi}\|_{L^1(\mu)} \right] \leq C_{R,d,M} n^{-\frac{1}{8}}$$

- 1 Introduction
- 2 Construction of the estimator
- 3 Convergence of our estimator
- 4 Discussion**

Prior results

- Many estimators have been proposed: 1NN, Kernel based, barycentric projection, entropic regularization... (Rigollet, Hutter, Manole, Divol, Niles-Weed...)

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$$\bar{\varphi} \text{ is } C^2, \quad \frac{1}{\lambda} I \preceq \nabla^2 \bar{\varphi} \preceq \lambda I, \quad \lambda > 0.$$

Caffarelli's regularity: known sufficient condition.

- 1 $\text{supp } \mu, \text{ supp } \nu$ convex compact, smooth boundary.
- 2 Densities bounded away from 0 and $+\infty$.
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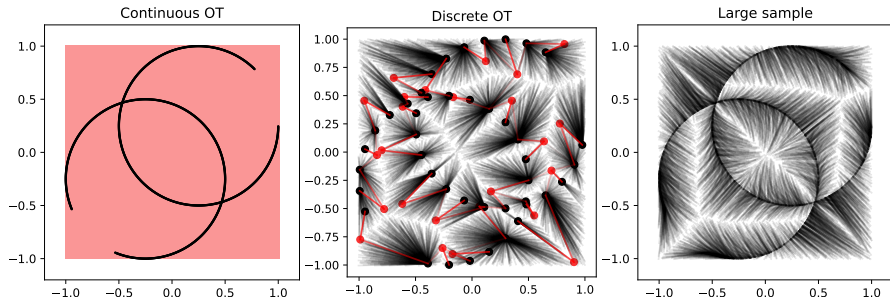
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In comparison, our results:

- Retain a similar statistical rate ($n^{-\frac{1}{4d}}$ vs $n^{-\frac{1}{d}}$).
- Don't require the Monge-diffeomorphism assumption, not even connectedness, only compact support and upper bounded density.

Visualisation



Uniform measure on the square is sent on the black measure.

$\hat{T}_{n,m}$ is computed from the empirical problem (red and black points). The estimator sends samples of the source to samples of the target (black lines).

High sample size.

Theorem

$$\mathbb{E} \left[\left\| \hat{T}_{n,m} - \nabla \bar{\varphi} \right\|_{L^1(\mu)} \right] \leq C_{R,d,M} \left(n^{-\frac{1}{4d}} + m^{-\frac{1}{4d}} \right) \quad (\text{General})$$

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- Proposed a simple and natural statistical estimator for transport maps.

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Future work will aim at proving that these rates adapt to the Monge diffeomorphism assumption and that they are minmax under the general assumptions we consider.

An improvement for our main Theorem using a novel result by Q.Mérigot

Shortly after our preprint, Quentin Mérigot proposed a novel stability bound similar to ours¹. Let $\varphi_{\mu,\nu}$ and $\varphi_{\mu',\nu'}$ be optimal Brenier potentials for the transport between μ, ν , resp. μ', ν' . Then for all $\epsilon > 0$:

$$\|\nabla\varphi_{\mu,\nu} - \nabla\varphi_{\mu',\nu'}\|_{L^1(\mu)} \leq C_{\epsilon,R,d} (W_1(\mu, \mu') + W_1(\nu, \nu'))^{\frac{1}{3+\epsilon}}.$$

This bound is near optimal in terms of exponent without further assumptions.

Using this stability result yields an improvement of our rate to $O(n^{-1/(3+\epsilon)d})$.

¹Q.Merigot, Sharp stability of Brenier maps via quantitative regularity of potentials